

PLASTICE

New technologies to integrate PLASTIC waste in the Circular Economy

Training materials

Module 2: Testing the technologies

Unit 2.3: Enzymatic hydrolysis: demonstrating mixed textile-waste recycling at pilot scale



This project has received funding from the European Union's Horizon Europe research and innovation programme under grant agreement n° 101058540.

MODULE 2 Overview

This module is composed of four units, each showcasing innovative approaches to plastic and textile waste recovery. The units present real-world demonstrations of how emerging technologies can support a circular economy through scalable and sustainable solutions. These include:

- Unit 2.1: Demo-sites for gasification and chemical post-treatment: enabling innovation in circular plastic technologies
- Unit 2.2: Pilot-scale plant for Microwave-Assisted pyrolysis of plastics for oil obtention
- **Unit 2.3: Enzymatic hydrolysis: demonstrating mixed textile-waste recycling at pilot scale**
- Unit 2.4: Demo-sites as innovation hubs for startups and academic entrepreneurs

These technologies have been developed and validated within the framework of the **PLASTICE project**, funded by the European Commission, which aims to promote circular solutions for plastic waste valorisation through **chemical recycling**.



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Unit 2.3

Enzymatic hydrolysis: Demonstrating mixed textile-waste recycling at pilot scale

*The enzymatic hydrolysis process described in Module 1 will be demonstrated at pilot scale within PLASTICE project. This includes a stepwise scaling up of the process from lab to pilot scale and the execution of a demo run, covering all steps from **textile sorting**, to the **chemical processes** to separate the fibres and recover the materials, to **yarn spinning** and fabric production. The entire consortium entrusted with the topic of textile recycling is working together on the demonstration of enzymatic recycling textile waste.*

Key points of Unit 2.3

- **5.8 million tons** of textile waste are discarded annually in the EU, with most ending up in landfills or incinerators.
- **Only 1%** of textile waste is recycled back into textiles; **25% is reused or downcycled**.
- Modern clothing blends **multiple fibres (e.g., cotton, polyester, wool, silk, synthetics)**, making recycling complex.
- Effective recycling requires **breaking down mixed fabrics** into their components.
- In PLASTICE, **enzymatic degradation** of cotton/polyester blends is used to recover polyester.



Sorting process

Recycling starts with the sorting process of discarded textiles. Those textiles, which can no longer be reused, are suitable for recovery in a recycling process. As textiles can be made of various fibres, the actual composition is not always clear. The technology we apply in PLASTICE to target and sort out the waste stream we are interested in, is near infrared spectroscopy (NIR). NIR uses infrared light, that is either reflected or absorbed by a sample, depending on its chemical structure. The data are then converted into a spectrum, revealing the composition of the measured textile. In this way, cotton and polyester fibres can be differentiated due to their different chemical structure.

As in our technology we dissolve the cotton part of a garment and recover the polyester fibres, we target a waste stream made of these two fibre types, preferably with a polyester share higher than 50% to keep the process efficient.

The sorting technology is demonstrated at DKW.

Preparation process

The textile waste cannot be subjected to our **enzymatic hydrolysis process without previous crushing**. This can be done with a cutting mill. Cutting the textile garments is also important to **remove hard parts**, that are not actual part of the fabric, but appear a lot in clothings like buttons, zippers and applications. These hard parts are removed by **metal separation or wind sifting**.

TCKT is equipped to demonstrate this preparation step of cutting the textiles at small pilot and pilot scale.

The **alkaline pretreatment** is part of the preparation process, where the shredded textiles are subjected to an alkaline solution, making the cellulose fibres more susceptible to the consequent enzymatic treatment, without harming the polyester.

The alkaline pretreatment is demonstrated at TCKT (small pilot scale) and AUSTRO (large pilot run).

Enzymatic recycling process

The alkaline pretreatment is directly followed by the actual enzymatic recycling process, the enzymatic hydrolysis of cotton. Hydrolysis is the **splitting of a chemical compound** by reaction with water. This is what happens to cotton, when it is broken down to **glucose**. This reaction applies



enzymes, which are proteins that work as catalysts for biochemical reactions. Catalysts accelerate reactions, without being consumed themselves and are necessary to achieve reasonable reaction times.

For **mixed fabrics**, this means that the cotton fibres are degraded into **sugar**, which is – in comparison to cotton – soluble in water. The **polyester fibres** are not affected by this reaction and remain as they are. The sugar solution can then be filtered off. The remaining polyester fibres are then washed and dried, before they can be further processed in the recycling process.

The **enzymatic hydrolysis** process is demonstrated at TCKT (small pilot scale) and AUSTRO (large pilot run).

Close the loop

The recovered polyester fibres can be extruded to **granules**. These recycled polyester granules are tested, if they achieve the required properties for further processing into fibres again.

TCKT and CTB have the plastic processing possibilities and the testing machines to demonstrate the conversion of the recovered polyester fibres to **spinnable granules** again.

Fibre spinning of the recycled granules, yarn production and consequent fabric production close the loop in fibre-to-fibre recycling, from waste to textile clothing again.

Fibre spinning, yarn and fabric production, as well as all related tests to these process steps, are done at small pilot and pilot scale at KORTEKS and SUN.

And also the **sugar solution** is not discarded, but can be reused as feedstock for the production of, for instance, bio ethanol.

Bioethanol production is not part of PLASTICE, but could be demonstrated at AUSTRO.



Conclusions

The complexity of textiles in recycling can be overcome with different separation strategies. These need to be energy efficient, which the shown enzymatic degradation is. With such a step, the textile mixtures can be decomposed, therefore enabling the recycling of the different components. Such processes need to be implemented industrially, but in a way that the majority of the streams are utilized – either for materials or as base chemicals for other applications. Overall, the approach of separating the components by means of a low energy approach is a green way to solve an until now unsolved issue of recycling of mixed textile waste.

In PLASTICE, several partners work together on the enzymatic degradation of the cellulose content in polyester/cotton textiles and demonstrate a closed fibre-to-fibre recycling process at pilot scale.



Bibliography

[1] Juanga-Labayen, J.P.; Labayen, I.V.; Yuan, Q. A Review on Textile Recycling Practices and Challenges. Textiles, 2022, 2, 174–188.

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NEXT

UNIT 2.4

Demo-sites as innovation hubs for startups and academic entrepreneurs



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